Retrieval-Augmented Generation Handbook

Page one introduces the handbook. Retrieval-Augmented Generation, or RAG, combines a retrieval system with a language model. Instead of relying only on what the model memorised during training, a RAG system first looks up relevant passages in a private knowledge base and then asks the model to answer using those passages. This reduces hallucination and allows answers to be traced to their sources.

Embeddings and Vector Databases

Page two explains how retrieval works. Each document chunk is converted into an embedding vector by an embedding model. The vectors are stored in a vector database such as FAISS, which can quickly find the nearest vectors to a query. Typical chunk sizes are between 500 and 1000 characters with a small overlap so that context is not lost at chunk boundaries. Dr. Ananya Rao from Bengaluru reviewed this design.

Conclusion

Page three concludes the handbook. The authors conclude that RAG is the most practical way to build trustworthy question answering over private documents. They recommend evaluating retrieval precision, answer correctness and citation accuracy, and they advise always showing source citations to the user.